KREISEL

# Technical Documentation

## Kreisel Electric KE_170589976_V3R1- Prototype Battery

Author: Peter Klepatsch

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**Kreisel Electric GmbH**
Promenade 25, A-4240 Freistadt

+43 (0) 7942 / 20404,
info@kreiselelectric.com

www.kreiselelectric.com

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# Technical specifications

| | |
|---|---|
| Battery Type | KE_170589976_V3R1 |
| Cell Technology | Li-Ion |
| Cell Manufacturer | Samsung |
| Cell Model | INR18650-29E |
| No. of cells in series | 90 |
| No. of cells in parallel | 21 |
| Nominal voltage | 324,0 V |
| Upper cut-off limit $U_{U_limit}$ | 378,0 V |
| Lower cut-off limit $U_{L_limit}$ | 225,0 V |
| Max. operating voltage $U_{max}$ | 373,5 V |
| - charging Voltage standard (extended range) | 373,5 V |
| - charging Voltage standard (daily use) | 360,0 V |
| Min. operating voltage $U_{min}$ | 270,0 V |
| Storage and delivery voltage ( ≤30% SOC) | 318,0 V |
| Capacity typical | 60,9 Ah |
| Energy content | 19,7 kWh |
| Max. discharge current continious (@25°C) | 126,0 A |
| Max. discharge current 10sec (@25°C) | 240,0 A |
| Max. charge current continious (@25°C) | 21,0 A |
| Max. charge current rapide (@25°C) | 42,0 A |
| Fuse rating (HV) | 100 A |
| Weight cells | 90,7 kg |
| Weight battery pack | 134 kg |
| Protection class | 0 |
| Operating temperature discharge | -20°C to 60°C |
| Operating temperature charge | -10°C to 45°C |
| Storage Temperature | 1 year : -20~25°C<br>3 months : -20~45°C<br>1 month : -20~60°C |
| Overall dimension (LxWxH) | 1020 mm x 610 mm x 240 mm |

# Terms of use

## Ambient and Operating temperature

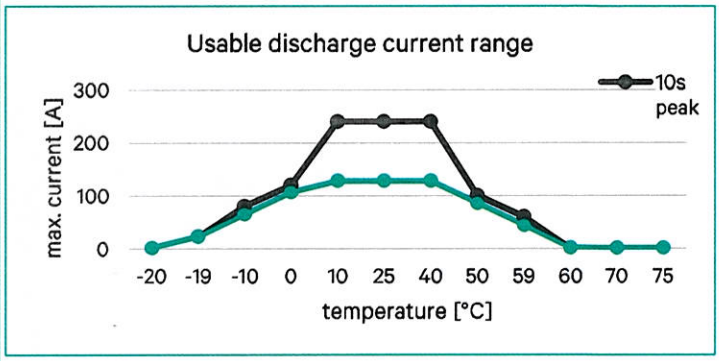
The monitoring of cell temperature carried out by thermistors. For a specific measurement and analysis of temperature, it is necessary that the battery pack is filled with enough cooling liquid (dielectric fluid).

The here described battery pack has a permissible operating temperature (discharging) range of -20 degrees Celsius to +60 degrees Celsius. An operation over or under this temperature can lead to restrictions in the life or in the worst case serious damage of the cells. To avoid this, a thermal shutdown has been integrated in the battery pack which turns off the system if the battery temperature gets higher than +65 degrees Celsius. The system will be shut down till the battery reaches a temperature of +40 degrees Celsius.

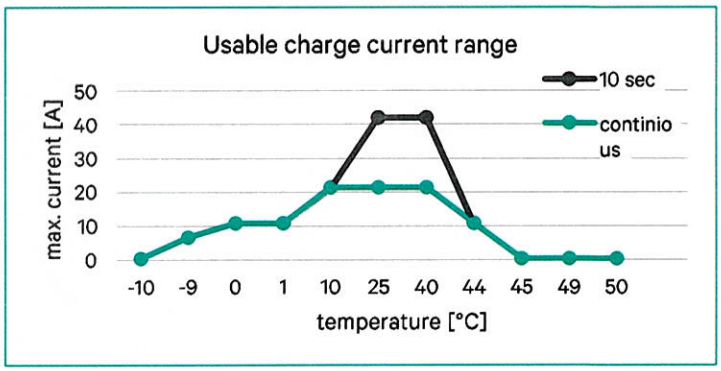
The temperature limits for (charging) the battery packs is set at 0 degrees Celsius to 40 degrees Celsius.

Between these temperature limits there are different discharge/charging power permitted as shown in the table below. Deviations from this values can cause major damage.

| Cell Surface | Max. discharge current | |
|---|---|---|
| Temperature [°C] | continious [A] | 10sec [A] |
| -20 | 0 | 0 |
| -19 | 21 | 21 |
| -10 | 63 | 80 |
| 0 | 105 | 120 |
| 10 | 126 | 240 |
| 25 | 126 | 240 |
| 40 | 126 | 240 |
| 50 | 84 | 100 |
| 59 | 42 | 60 |




| Cell Surface | Max. charge current | |
|---|---|---|
| Temperature [°C] | continious [A] | 10 sec [A] |
| -10 | 0 | 0 |
| -9 | 6 | 6 |
| 0 | 11 | 11 |
| 1 | 11 | 11 |
| 10 | 21 | 21 |
| 25 | 21 | 42 |
| 40 | 21 | 42 |
| 44 | 11 | 11 |



## Voltage

The nominal voltage results from the number of cells being connected in series. For this battery pack the nominal voltage is 324DC (90 x 3.6V).

The max. operation voltage is limited to 374 VDC (4,15V cell voltage). The min. operation voltage is limited to 270 (3,0V cell voltage).

The voltage limitation (Upper-/ Lower cut-off voltage) is controlled by the battery management system. The safety limits will shut off the system if the voltage is above 378 VDC (4,2V cell voltage) and or lower than 225 VDC (2,5V cell voltage).

To avoid any system shut-downs we recommend to reduce the power load especially in the low voltage area, so that the voltage (under load) can never drop below the lower cut-off limit. The Voltage limits, SOC and power modes we recommend are highlighted in the table below.

To extend the lifecycle of the battery it is recommend to charge the battery to 360 VDC for daily use. If a longer driving range is required then of course the battery system can be charged to a higher voltage level. In that case the extend rage voltage is limited to 374VDC. So the extend range voltage is equal to the max. operating voltage. Any values above this limit can cause damages to the battery.

| SOC | Voltage cell @25°C [V] | Power mode (discharge) |
|---|---|---|
| 100% | 4,149 | 100% |
| 95% | 4,106 | 100% |
| 90% | 4,076 | 100% |
| 85% | 4,05 | 100% |
| 80% | 3,99 | 100% |
| 75% | 3,941 | 100% |
| 70% | 3,899 | 100% |
| 65% | 3,858 | 100% |
| 60% | 3,81 | 100% |
| 55% | 3,755 | 100% |
| 50% | 3,71 | 100% |
| 45% | 3,647 | 100% |
| 40% | 3,646 | 100% |
| 35% | 3,622 | 100% |
| 30% | 3,599 | 100% |
| 25% | 3,567 | 100% |
| 20% | 3,517 | 100% |
| 15% | 3,47 | 100% |
| 10% | 3,427 | 100% |
| 5% | 3,299 | 50% |
| 0% | 3,022 | 50% |
| | 3 | 0% |

## Current (power modes)

In short, the discharge and charging current (power) is determined by the temperature and voltage (SOC) of the battery. The maximum discharging current of the battery pack is 200Ampere. But as mentioned above this discharge current does not apply to the whole discharge cycle.

The battery pack is also fitted with an overcurrent safety mechanism to protect the battery against high current abuse. For this battery the discharging limit is set to 252 Ampere. Any value above this limit will shut off the battery system. Also in charging mode the system will shut off if the current fed to the battery is higher than 80 Ampere.

To avoid system damages and failures please make sure that these current limits are set in your power electronics (Inverter/Charger).

## Peak Current Demand

The following formula shall be used in case the systems peak discharge capability is required.

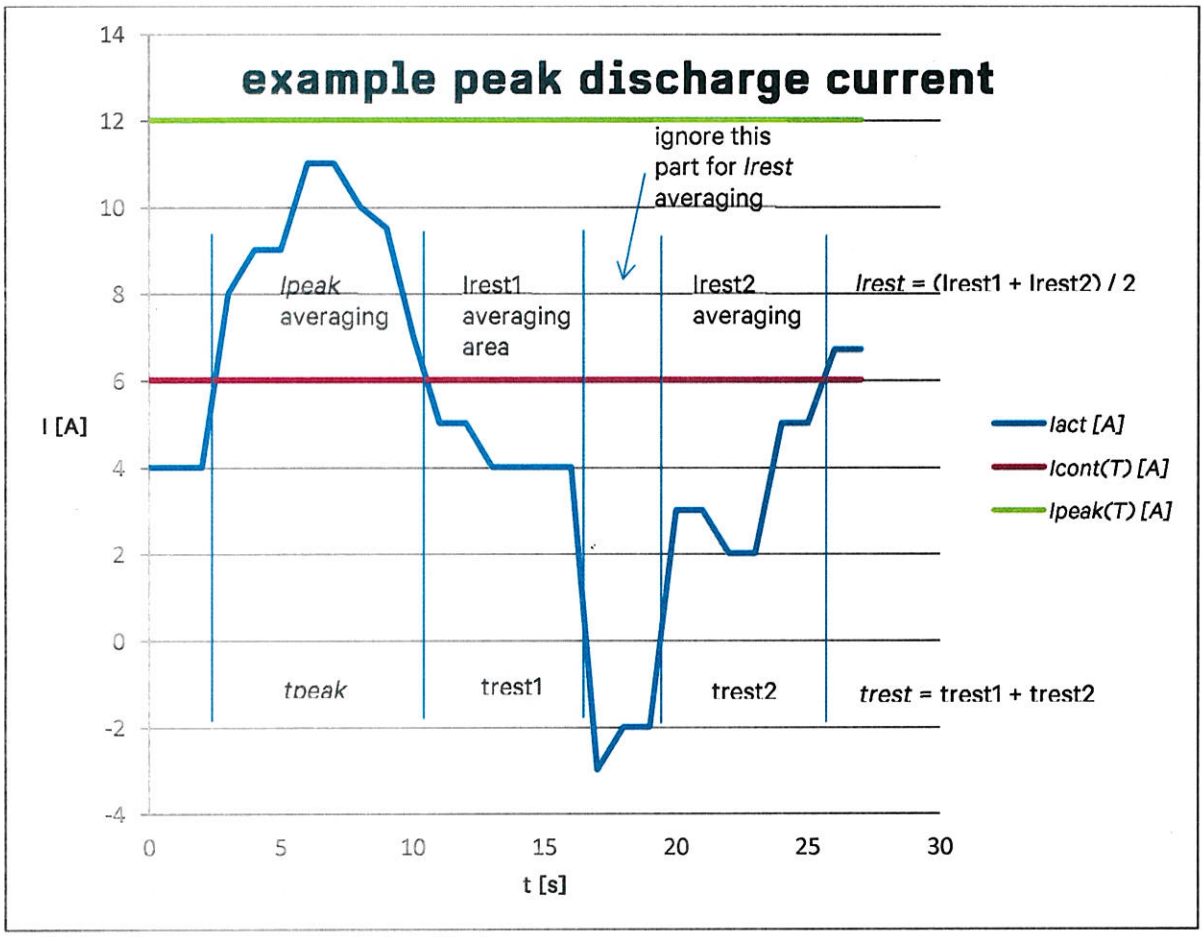
$Ipeak$: Average value of demanded peak discharge current (before the resting phase)
$Irest$: Average value of resting current after a current peak
$Icont(T)$: Maximum continuous discharge current dependent on temperature (see above)
$Ipeak(T)$: Maximum peak discharge current dependent on temperature (see above)
$tpeak$: Time of the current peak
$trest$: Required resting time (at Irest) before a subsequent current peak is allowed

$$\mathrm{Irest} = \frac{\mathrm{Icont(T)} * (\mathrm{tpeak} + \mathrm{trest}) - \mathrm{Ipeak} * \mathrm{tpeak}}{\mathrm{trest}}$$

$$trest = -\frac{(Ipeak - Icont(T)) * tpeak}{Irest - Icont(T)}$$

The formula shall be used within the following boundaries:

$$0s < tpeak \leq 10s$$
$$0s < trest$$
$$Icont(T) < Ipeak \leq Ipeak(T)$$
$$0A \leq Irest < Icont(T)$$

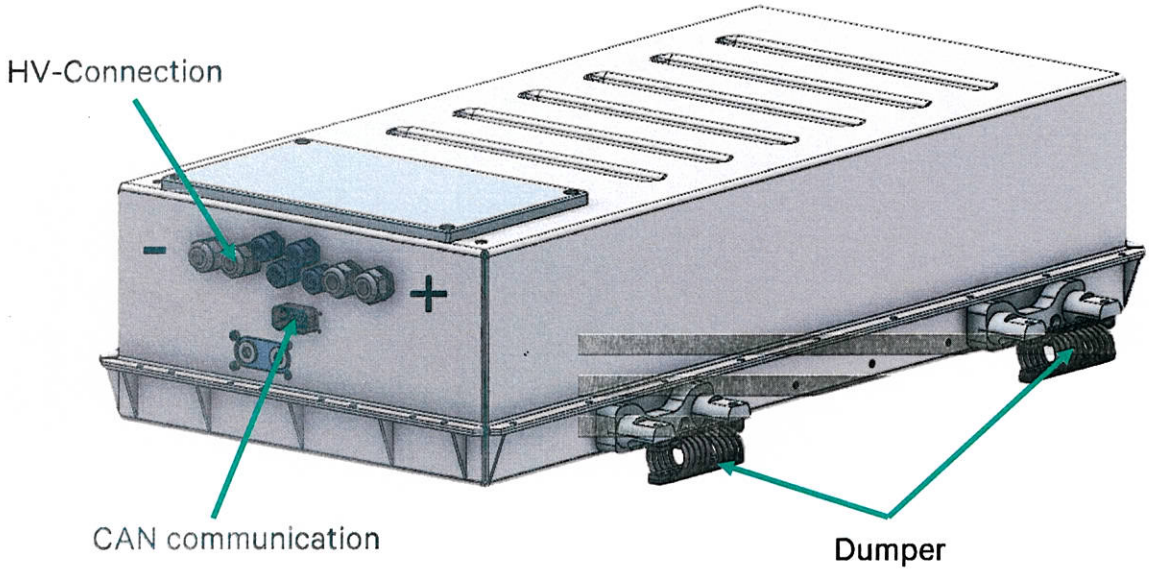



A peak discharge current for a maximum duration of 10 seconds can be demanded at any time as long as a resting phase according to the formula above is achieved.

If this requirement is not met properly the battery temperature may rise uncontrollable. The BMS will shut off the main contactors at the specified maximum temperature.

# Interfaces/Control

## Battery interfaces




## Communication and Start up procedure (STW mBMS)

The communication is carried out via the standard CAN 2.0A bus system and uses a Bit rate of 500kBaud. Details see in appendix. (STW_mBMS.zip)

The control dbc-file contains the CAN-ID, offset and factor of every necessary Parameter. The sensor dbc-file contains more detailed battery data, which shouldn't be used normally. They can be opened with every Text-Editor. But for more convenient viewing the "Vector CANalyzer Demo" (http://vector.com/vi_canalyzer_de.html ) is recommended. It contains a Database viewer for CAN dbc-files (CANdb.exe).

Also a description of the start up procedure for the mBMS in the form of a flow chart is included.

Mainly it is necessary to send the CAN Message with ID 0x7FF to the mBMS. There are 2 main values inside this massage to be considered:
Func_sys_state_req: always send the value 1 as soon as you start up the CAN Communication with the mBMS.
Func_MSW_Request_Master: send the value 0 to open; and the value 1 to close the main contactors of the battery.

After that all the information of the current battery status are sent on the CAN Bus, like Temperature, SOC, Warnings, Errors, Single Cell voltages and many more, which are described in the dbc-File.

The mBMS also sends the battery limits (like Current, Voltage, Temperature, State of Charge...). It is mandatory to stay within the limits, otherwise serious damage of the battery might occur.

These controls are required to start the battery:

- T15 : You shall not connect this port, if you want to control the mBMS by CAN Bus.
- T30 (fuse 8A)
- T30c (fuse 20A)
- GND

# Mounting

The battery pack must be mounted at dumper (shock absorber) "Sebert J6 54-10".
For a fully function and durability of battery pack and dumper, a professional mounting is strongly requested.

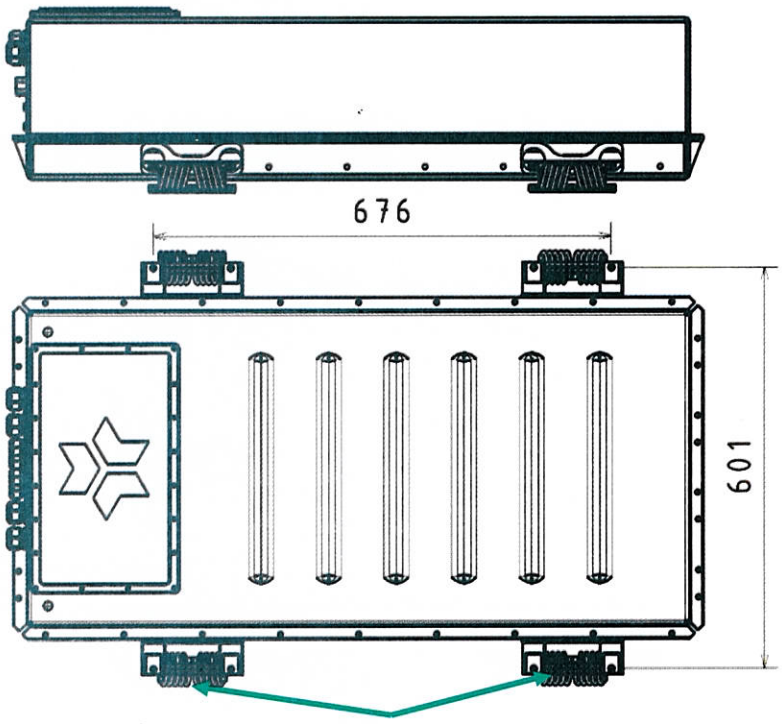
Make sure that all dumpers are mounted without bracing ore friction,
the mounting points hast to be aligned.

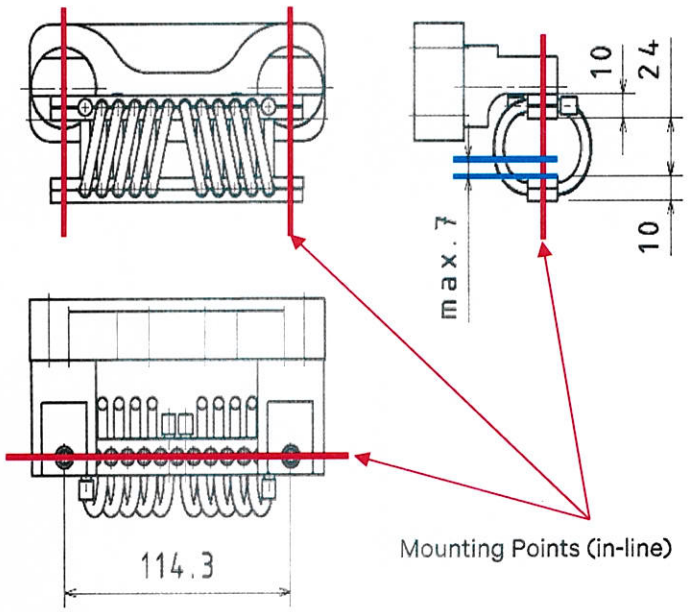
The dumper isn't suitable to compensate differences at the mounting points.

Also make sure the necessary clearance of dumping height and use flat had screws with max 7mm head.

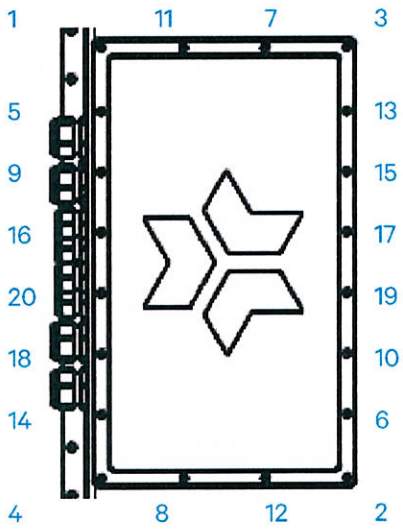
In combination with our battery pack, the dumper is configurated for 5 times static overload (5 G)

Tighten bolts of the maintenance cover diagonally with a torque of 2Nm.




Tightening sequence for the
Maintenance cover (2Nm):



# Delivery status and deliveries

The battery system will be delivered with a cell voltage of about 3,6V

The whole battery system consist of following parts:

- Battery pack
- Shock absorber
- Documentation

# Appendix

- Introduction_INR18650-29E_120308
- STW_mBMS_Control_CAN
- Datasheet Shock absorber J6 54-10
- PIN Assignment